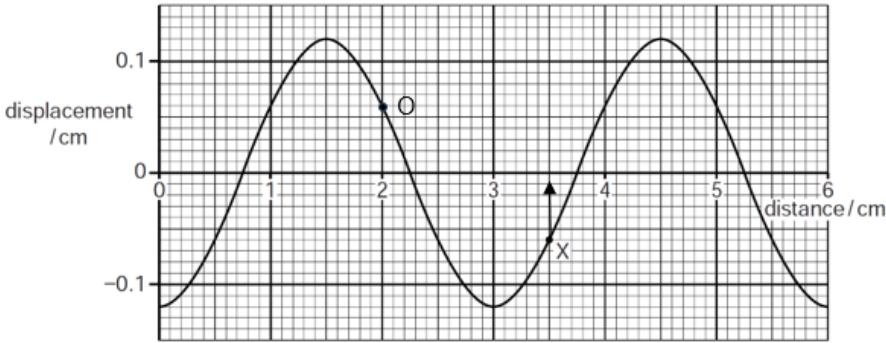


Section B

7(a)(i)	O is on the line in an adjacent segment of X, that is not a node.	A1
CKW MC	This part was well done	
7(a)(ii)	arrow drawn at X pointing vertically upwards 	A1
CKW MC	This part was generally well done, although there were answers in other directions.	
7(a)(iii)	distance = $2 \times 0.06 \text{ cm} = 0.120 \text{ cm}$	A1
CKW MC	This part was well done, though some candidates did not realise the distance travelled was for half a period.	

7(b)	$\frac{1}{2}\lambda = 65.0 \text{ cm}$ $\text{speed} = 0.650 \times 2 \times 256$ $= 333 \text{ m s}^{-1}$	M1 A1
CKW MC	Some candidates did not deduce the correct wavelength.	
7(c)(i)	$b \sin \theta = \lambda$ $\sin \theta = \frac{633 \times 10^{-9}}{0.0800 \times 10^{-3}}$ $\theta = 0.453^\circ$ $w = 2 \times 5.12 \times \tan \theta = 2 \times 5.12 \times \tan (0.453^\circ)$ $= 0.0810 \text{ m}$	C1 A1
CKW MC	This part was generally well done. Usually, the wrong answer was off by a factor of 2.	
7(c)(ii)1	Destructive interference occurs when light from second slit meets	

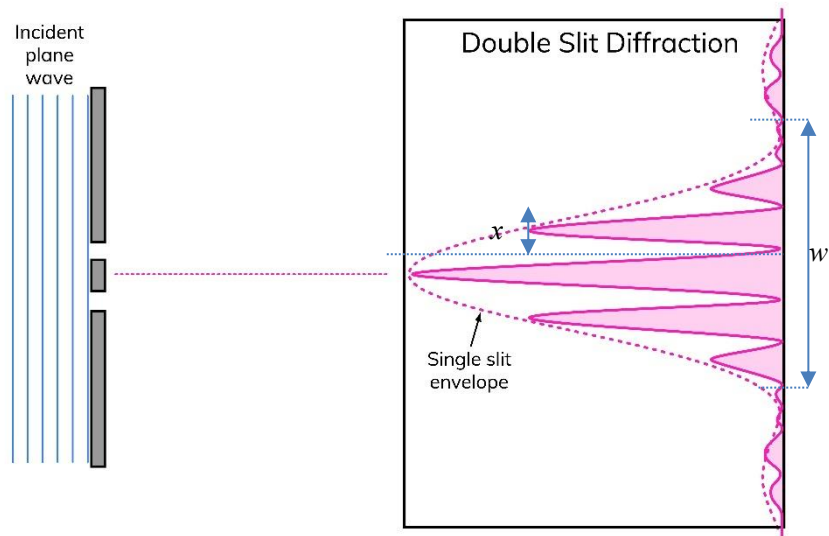
7(c)(ii)1.	Destructive interference occurs when light from second slit meets anti-phase with light from first slit.	B1
CKW MC	This part was well done	
7(c)(ii)2.	$x = \frac{\lambda D}{a} = \frac{(633 \times 10^{-9})(5.12)}{0.240 \times 10^{-3}}$ $= 0.0135 \text{ m}$	C1 A1
CKW MC	This part was well done	
7(c)(ii)3	$w = 6x \approx 0.0810 \text{ m}$ (The 3rd maxima from the central maxima of the double-slit interference pattern coincides with the 1st order minima of the single slit spectrum due to the width of the slit.) These places are minima of the single slit envelope.	B1 B1

CKW
MC

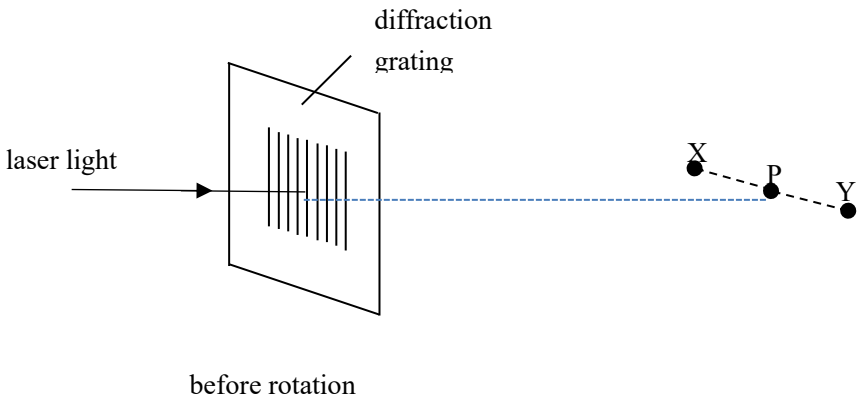
Realistic Double Slits

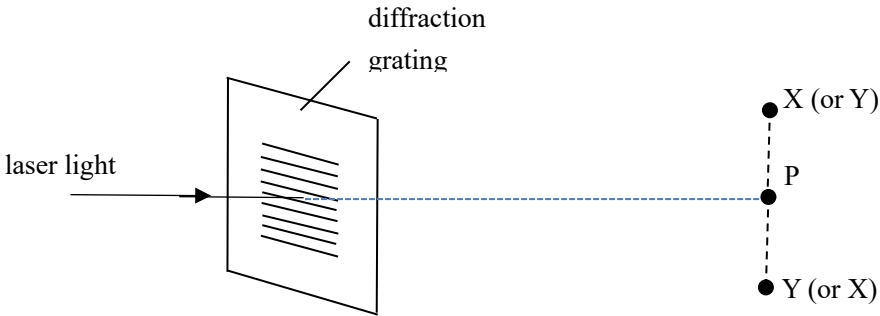
In real life, the double slits are never infinitely thin (they are not really point sources), they actually have a width.

- This means that each of the two slits will generate a single-slit diffraction pattern.
- The result is that we have a **single-slit envelope** with the pattern of dark and bright fringes from the actual double slit interference.



- To get information about the **distance** between the two slits, you have to look at maxima generated by the **double slit pattern**.
- To get information about the **width** of each slit, you have to look at the **position of the 1st order minima** of the single slit envelope.

7(d)(i)	$\tan \theta = \frac{38}{165} = 0.230$ or $\sin \theta = \frac{38}{\sqrt{38^2 + 165^2}} = 0.224$ or $\theta = 12.97^\circ$ $d \sin \theta = \lambda$ $d \sin 13^\circ = 6.33 \times 10^{-7}$ $d = 2.82 \times 10^{-6} \text{ m}$ number per metre = $\frac{1}{d}$ $= 3.5 \times 10^5 \text{ m}^{-1}$	C1 C1 A1
CKW MC	- Most candidates omitted the unit. - Small angle approximation is not needed. $\theta = \frac{s}{r} = \frac{38}{165} = 0.230 = \tan \theta \neq \sin \theta$	
7(d)(ii)	1 Lines are further apart in second order, than in first order. 2 Lines are brighter and sharper in first order, than in second order.	B1 B1
CKW MC	- Most candidates did not provide a complete comparison of the shape of the maxima i.e. both the intensity and width. - The wavelengths of 633 nm and 638 nm are too close for the colours to be different.	
7(e)(i)	P remains in the same position. X and Y rotate through 90°.	B1 B1
 laser light diffraction grating X P Y before rotation		

 after rotation		
CKW MC	- It is not possible to deduce if the rotation is clockwise or anticlockwise. - It is important to identify that the position of P is not affected, as it is possible that there is a translation in the diffraction pattern.	
7(e)(ii)	screen is not parallel to grating OR grating is not normal to incident light	B1
CKW MC	Many candidates did not realise the same grating was used, so the adjacent slit centre-to-centre separation remains unchanged.	